Technical report
Sparse Principal Component Solver

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May 22, 2026

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0.1 Sparse Principal Component Problem

A formulation of the Sparse Principal Component Problem is as follows

$$(\text{SPCA}_{\Sigma}^{k}) \begin{cases} \max & x^T \Sigma x \\ s.t. & x^T x = 1, \\ & \|x\|_0 \leq k, \\ & x \in \mathbb{R}^n, \end{cases} \tag{1}$$

where the integer parameter $1 \leq k \leq n$ controls the sparsity of the solution.

0.2 Extracting several sparse components

A well know technique to extract more than one sparse component is by deflating the covariance matrix at each round. see Mackey et al.

Algorithm 1 Generalized Deflation Method for Sparse PCA

Given: covariance matrix Σ_0, number of rounds r, $\{k_1, ..., k_r\} \subset \{1, ..., n\}$.
$B_0 := I$,
for $t = 1, ..., r$ **do**
  $x_t := \underset{x^T B_{t-1} x = 1, \|x\|_0 \leq k}{\text{argmax}} x^T \Sigma_{t-1} x$
  $q_t := B_{t-1} x_t$
  $\Sigma_t := (I - q_t^T q_t) \Sigma_{t-1} (I - q_t^T q_t)$
  $B_t := B_{t-1}(I - q_t^T q_t)$
  $x_t := x_t / \|x_t\|_2$
end for
return $\{x_1, ..., x_r\}$

An approach for solving point 4: in Algorithm 1 is to first replace $x^T B_{t-1} x = 1$ by $x^T (B_{t-1} + \epsilon I) x = 1$ and use Woodbury matrix identity to transform it in the form of 1. To be detailed!

0.3 Implemented methods

- Dca method from Thiao et al.
- Gspca forward and backward methods from Moghaddam et al.
- Mav and amvl, iterative elimination methods from Wang et al.
- We will implement all the sparse principal component methods and deflation techniques and incorporate them into the solver.

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